# Thermodynamics and Statistical physics
# 1st Test

## I. THEORY QUESTIONS

1. State the Zeroth law of thermodynamics. How is it related to the definition of temperature?

2. When is the formula $đW = -PdV$ for infinitesimal work applicable?

3. What is an adiabatic process? Give a definition of internal energy in terms of such processes.

4. An isolated system is divided into two halves by a wall in the middle. Both sides of the wall contain a gas of the same molecule. What are the conditions for thermal equilibrium between the two halves if the wall is
   a) Diathermal but immovable and impermeable?
   b) Diathermal and movable but impermeable?
   c) Diathermal, movable and permeable?

5. Demonstrate the equivalence of Kelvin's and Clausius's statements of the Second law of thermodynamics.

6. State the fundamental thermodynamic identity that combines the First and Second laws in the form $dU = \ldots$. For what processes is it applicable?

7. What is the condition for thermal equilibrium in a system in thermal contact with an environment at a fixed temperature $T$? Discuss what happens in the limits $T \to 0$ and $T \to \infty$.

8. Consider the following chemical reaction:

$$2\ H_2O \rightleftharpoons 2\ H_2 + O_2.$$

   Write the condition for chemical equilibrium between the three molecules taking part in the reaction.

## II. PROBLEMS TO SOLVE

1. A railway track is made of steel rails, each 15 m long, initially laid at a temperature of 5 °C. During summer, the temperature can rise to 50 °C. If the rails are not given enough space to expand, they will exert a compressive force on each other, leading to dangerous buckling (also called *sun kinks*).

   The coefficient of linear expansion for steel is $\alpha = 1.2 \times 10^{-5}$ 1/K, and the Young's modulus for steel is $E = 2.0 \times 10^{11}$ Pa. The rails are rigidly fixed, meaning they cannot expand freely.

   (a) Calculate the expansion of a single 15 m rail when the temperature increases from 5 °C to 50 °C.

   (b) If the rail cannot expand, it will experience thermal stress. Determine the stress developed in the rail. *Hint: The stress under compression of a rod is* $\sigma = E\Delta l/l$.

   (c) If the cross-sectional area of the rail is 50 $\text{cm}^2$, calculate the force that each rail segment exerts on its surroundings.

   (d) Railway engineers leave small expansion gaps between rails to prevent dangerous buckling. If the maximum allowable force before buckling occurs is $1.5 \times 10^6$ N, determine the minimum gap that should be left between each 15 m rail to prevent excessive stress. Assume that the temperature of the railway does not go beyond 50 °C.

   ***Solution***

   (a) $\Delta T$ =45 K, $\Delta\ell = \alpha\ell\Delta T = 0.81$ cm

   (b) $\sigma = E\Delta\ell/\ell = E\alpha\Delta T = 1.1 \cdot 10^8$ Pa,

   (c) $F = \sigma A = 5.4 \cdot 10^5$ N

   (d) $F = EA\alpha\Delta T - EAg/\ell \implies g = (EA\alpha\Delta T - F^*)\ell/E/A < 0$. Therefore no gap is sufficient.

2. An ideal gas heat engine operates using one mole of an ideal gas in a Brayton cycle, which consists of the following four stages:

   - Adiabatic compression: The gas is compressed from state 1 to 2 with no heat exchange (work input required).
   - Isobaric heat addition: The gas absorbs heat at constant pressure from state 2 to 3, performing work.
   - Adiabatic expansion: The gas expands from state 3 to 4, producing work with no heat exchange.
   - Isobaric heat rejection: The gas releases heat at constant pressure from state 4 back to 1.

   The engine operates between a low temperature $T_1 = 300$ K and a high temperature $T_3 = 1200$ K. The volume ratio between the states is $V_1/V_2 = 5$, and the gas follows the ideal gas law with a heat capacity ratio $\gamma = 1.4$.

   (a) Derive an expression for the final temperature $T_2$ after adiabatic compression. Compute the work done on the gas during this process.

   (b) Derive an expression for the heat $Q_{\text{in}}$ added to the gas during the isobaric expansion. Compute $Q_{\text{in}}$ using the given temperatures.

   (c) Compute the total work output of the cycle and determine its efficiency. Compare the efficiency with the Carnot efficiency operating between the same temperature limits.

   ***Solution***

   (a) For an adiabat: $TV^{\gamma-1} = const$, therefore

   $$T_2 = T_1 \frac{V_1^{\gamma-1}}{V_2^{\gamma-1}} = T_1 5^{\gamma-1} = 571\text{K}$$

   and

   $$W = \Delta U = \frac{p_2V_2 - p_1V_1}{\gamma - 1} = nR\frac{\Delta T}{\gamma - 1} = 5.6\text{kJ}$$

(b)

$$Q_{\text{in}} = \Delta U - W = p\Delta V \frac{\gamma}{\gamma - 1} = nR\Delta T \frac{\gamma}{\gamma - 1} = 18.3\text{kJ}$$

(c)

$$T_4 V_4^{\gamma-1} = T_3 V_3^{\gamma-1}$$

and

$$V_4 = nRT_4/p_1 = V_1 \frac{T_4}{T_1}, \quad V_3 = nRT_3/p_2 = V_2 \frac{T_3}{T_2} \implies V_1^{\gamma-1} \frac{T_4^{\gamma}}{T_1^{\gamma-1}} = V_2^{\gamma-1} \frac{T_3^{\gamma}}{T_2^{\gamma-1}}$$

$$T_4 = \left( \frac{V_2 T_1}{V_1 T_2} \right)^{\frac{\gamma-1}{\gamma}} T_3 = 630\text{K}$$

(d) Work done by the gas in the second and third processes.

$$W_{2g} = p\Delta V = nR\Delta T = nR(T_3 - T_2) = 5.2\text{kJ}$$

$$W_{3g} = -nR \frac{\Delta T}{\gamma - 1} = 6.9\text{kJ}$$

So the full work output is $W_g$ =12.1 kJ. The efficiency is then $e = W_g/Q_{\text{in}} = 0.66$. The Carnot efficiency is $e_C = 0.75$.

3. Prove the Maxwell relation:

$$\left( \frac{\partial T}{\partial p} \right)_S = \left( \frac{\partial V}{\partial S} \right)_p .$$

***Solution*** $H = U + pV \implies \mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$

$$\left( \frac{\partial H}{\partial S} \right)_p = T, \left( \frac{\partial H}{\partial p} \right)_S = V \implies \left( \frac{\partial T}{\partial p} \right)_S = \left( \frac{\partial V}{\partial S} \right)_p$$